Mino Puzzle

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The Game of Tetris

Tetris is one of those games that transcends the wearing down of time—it seems it has never been more popular than now, played by nerdy Stuyvesant students in the halls and computer labs, despite its first version being developed in 1984.

Its history isn't terribly significant in any mathematical way, but its original creator, Alexey Pajitnov, originally had an alternative idea to the premise of the game. Though the fundamentals would have been the same—stacking, clearing lines, getting points—he had originally planned on **pentominoes** instead of the **tetrominoes** that we know of.

But... why would he *change* that facet of the game? Why not keep it as pentominoes? Sure, there's the obvious notion of decreasing piece count to not make the game overbearing, but surely there must have been something else. Maybe exploring **minos** could help one gain some insight on this.

It helps to define the following:

- A **mino** is a square cell.
- A **polyomino** is a connection of minos such that each is connected to at least one other mino by an edge.
- An n**-mino** is a polyomino composed of n minos (though **pentominoes**, made of 5 minos, and **tetrominoes**, made of 4 minos, will be referred to by these names).

The solution is on page 13.

Part I

1. Consider a 7x4 rectangle. Is it possible to fill this square using one of each tetromino (namely, the I, O, L, J, S, Z, and T piece)?
2. Consider placing a T-tetromino on the bottom of a limitless board. What placement of a follow T-tetromino would best accommodate the remaining pieces?
3. Is it possible to tile a 3x20 board using each pentomino exactly once?
4. Is it possible to tile two 6x5 rectangles using each pentomino exactly once?

Part II

1. Let $M(n)$ be the number of pieces of an n-mino. For example, $M(4) = 7$ because there are 7 distinct tetrominoes. Try to come up for an expression of $M(n)$ in terms of n.